Family	ID	Strategy	Inspiration	Direct messages	Shared medium	Memory	Hierarchy	Distance decay	Roles	Negative feedback	Quorum
baseline	S0a	Oracle	upper bound								
	S0b	Solo	lower bound								
abstract	S1	Independent + majority	voting								
	S2	Free debate	deliberation	■							
	S3	Particle swarm (PSO)	flocking, abstracted	■		■					
insects	S4	Ant-colony stigmergy	ant pheromone trails		■					evaporation	
	S5	Artificial bee colony	bee foraging		■	■			■	abandon	
	S9	Honeybee nest-site	bee nest-site choice		■				■	decay+stop	■
	S10	Response threshold	insect task roles		■				■	verify	
wolves	S7	Grey-wolf hierarchy	wolf-pack ranks	■			■		■		
fireflies	S6	Firefly attraction	firefly flashing	■				■			
birds	S8	Boids	bird flocking	■						separation	